Automatic Root Cause Analysis for LTE Networks Based on Unsupervised Techniques

Ana Gómez-Andrades, Pablo Muñoz, Inmaculada Serrano, and Raquel Barco

***Abstract*—The increase in the size and complexity of current cellular networks is complicating their operation and maintenance tasks. While the end-to-end user experience in terms of throughput and latency has been significantly improved, cellular networks have also become more prone to failures. In this context, mobile operators start to concentrate their efforts on creating self-healing networks, i.e., those networks capable of troubleshooting in an automatic way, making the network more reliable and reducing costs. In this paper, an automatic diagnosis system based on unsupervised techniques for Long-Term Evolution (LTE) networks is proposed. In particular, this system is built through an iterative process, using self-organizing maps (SOMs) and Ward's hierarchical method, to guarantee the quality of the solution. Furthermore, to obtain a number of relevant clusters and label them properly from a technical point of view, an approach based on the analysis of the statistical behavior of each cluster is proposed. Moreover, with the aim of increasing the accuracy of the system, a novel adjustment process is presented. It intends to refine the diagnosis solution provided by the traditional SOM according to the so-called silhouette index and the most similar cause on the basis of the minimum Xth percentile of all distances. The effectiveness of the developed diagnosis system is validated using real and simulated LTE data by analyzing its performance and comparing it with reference mechanisms.**

***Index Terms*—Diagnosis, fault identification, Long-Term Evolution (LTE), self-healing, self-organizing maps (SOMs), silhouette, unsupervised learning.**

I. Introduction

Cellular networks are an effective way of communication that allows people to instantaneously send and receive information anywhere. Over the past few years, as a consequence of a sharp increase in the traffic demand, the infrastructure of cellular networks has been profoundly modified. The higher complexity of these networks has encouraged mobile operators to implement effective and low-cost management algorithms. In this context, self-organizing networks (SONs) establish a new concept of network management in which the operation and maintenance tasks are carried out with a high level of automation [1]. Within this paradigm, self-healing is a major SON category whose aim is to automate the troubleshooting process [2], [3], which is composed of the detection, diagnosis, compensation, and recovery phases. As cellular networks are currently more prone to failures due to their huge increase in size and complexity, self-healing is gradually gaining attention from operators. In this sense, the enormous diversity of performance indicators, counters, configuration parameters, and alarms can be used to develop more intelligent and automatic techniques that cope with faults in a much more efficient manner. The implementation of these mechanisms has a direct impact on network availability and the operator's revenues.

The aim of this paper is to present an automatic diagnosis system that can be part of a self-healing Long-Term Evolution (LTE) network. Although there are some references related to automated diagnosis in the literature [4]–[6], it is extremely difficult to deploy those proposed systems in real networks. This is due to the fact that the design of these systems is based either on expert knowledge or on historical databases of fault cases. On the one hand, experts in troubleshooting do not have either the time or the expertise to build the required complex models. On the other hand, although there are historical databases of performance indicators, they do not contain any indication of whether there is a fault or its cause. Therefore, compared with previous references, the key characteristic of the proposed system is that it provides both a simple way of analyzing the behavior of faults and an accurate fault identification without requiring labeled historical databases (labeled means that the fault cause has been identified and that it is included in the database). In this context, there are two types of learning algorithms, namely, supervised and unsupervised, whose application depends on whether the training data set includes additional information about the cause of the problem or only the raw data (i.e., performance indicators), respectively.

This paper presents an automatic diagnosis system based on different unsupervised techniques with SOM as the centerpiece. In particular, the proposed system consists of three novel approaches.

- The first approach is the proposed unsupervised method to automatically identify the number of clusters that an expert will consider statistically different. To that end, the combined use of the Davies–Bouldin (DB) index and the Kolmogorov–Smirnov (KS) test [7] is proposed.
- The second approach is related to how expert knowledge is introduced in the system. To make the system autonomous during the exploitation stage, expert knowledge is

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included in the design stage. The proposed method focuses on the study of the statistical behavior of the clusters by inspection of the training data associated with each of them, instead of directly analyzing the characteristics of the neurons, which provides less detailed information. With this information, experts are able to identify whether each individual cluster is relevant and represents a particular fault of mobile networks. As a result, the diagnosis system will be able to diagnose a set of faults that has been validated and approved by experts.

- The third contribution is the proposed adjustment in the exploitation stage. Once the diagnosis system is built, it can be used as part of a self-healing system to automatically diagnose faulty cells, i.e., to determine the cause of the problem. As mentioned earlier, this diagnosis must be as accurate as possible. Therefore, the main difference with other SOM systems is that the mapping between the input metrics and the root cause includes an additional adjustment phase where the worst cases (i.e., those whose diagnosis is not clear) are analyzed in detail. In particular, the proposed approach intends to find out the cause with greatest similarities among all neighboring cases on the basis of the distances within each case. Then, this is compared with the initial decision corresponding to the original SOM algorithm, and the solution that maximizes the average silhouette index [8] is determined.

The rest of this paper is organized as follows. First, the existing work related to fault diagnosis in mobile communication systems is presented in Section II, followed by the motivation for the design of an unsupervised diagnosis system in Section III. Then, in Section IV, the proposed system is described in detail, explaining the concepts and the algorithms used in each stage. Section V demonstrates the validity and the robustness of this approach using real and simulated data and comparing its efficacy with reference mechanisms. Finally, in Section VI, the main conclusions are discussed.

II. Related Work

Although automatic fault identification is essential for the prompt enforcement of maintenance decisions, related literature in the field of mobile networks is scarce. Thus, it remains an open research problem. Nevertheless, some solutions for mobiles networks can be found in the literature. An overview in the field of mobile network is presented below.

- Bayesian networks: Some studies such as [4] and [5] proposed the use of a Bayesian network as classifiers of fault problems to automatically diagnose them. In particular, Barco *et al.* [4] presented a method based on a naïve Bayesian classifier to identify the fault cause in global system for mobile communications/general packet radio service, third-generation (3G), or multisystem networks using performance indicators as continuous variables. In addition, Khanafer [5] presented another automated fault diagnosis system based on Bayesian networks for Universal Mobile Telecommunications System networks using two different algorithms to discretize the performance indicators [i.e., percentile-based discretization (PBD) and entropy minimization discretization].
- Scoring-based systems: A different approach based on a scoring system has been proposed in [6]. This detection and diagnosis system is automatically built from the labeled fault cases reported by experts, and it uses a scoring system to determine how well a specific case matches each diagnosis target. Novaczki [9] proposed more sophisticated profiling and detection capabilities to improve this framework.

  All these techniques proposed in the literature, i.e., [4]–[6], [9], are supervised techniques. Therefore, their diagnosis process requires a historical database of labeled fault cases to learn the impact the faults have on the performance indicators. However, when the set of documented and solved cases is poor, which is partly due to the limited occurrences of each fault, unsupervised techniques are the most appropriate. This is precisely what occurs with the historical records of faults in mobile networks because experts are not concerned with storing the cause of the problem or the actions they took when performing their troubleshooting tasks.
- Neural networks: Several works have demonstrated the potential and utility of using the unsupervised neural network known as the self-organizing map (SOM) [10] to automate the fault detection phase of the troubleshooting tasks (that is, the phase prior to diagnosis). In particular, in [11], SOM is used to build the "normal profile" of the network and to determine the healthy ranges of the selected performance indicators (i.e., the symptoms). Therefore, this system helps identify whether the symptoms are healthy or faulty, without ever providing the fault cause. In addition, [12]–[15] show how SOM can be used to analyze multidimensional 3G network performance data to aid in the manual fault diagnosis. The aim is to cluster cells based on their performance to assist experts in their manual troubleshooting and parameter optimization tasks. In contrast, the system proposed in this paper aims to provide automatically and directly the fault cause of a problematic cell without any supervision of the experts in the exploitation stage. Therefore, it is essential for the proposed system to ensure two important aspects: 1) All identified clusters must present different statistical behaviors to guarantee that there are no similar clusters associated with the same fault cause, and 2) the final diagnosis must be as accurate as possible. Although the proposed system is based on the SOM technique, as previously stated, the purpose of the whole system is different from that presented in [12]–[15]; thus, the system proposed in this paper consists of additional and complex techniques to achieve those requirements.

Regarding the use of the silhouette index with SOM techniques, there are some reference in the literature (not related to wireless networks), such as [16] and [17]. However, in those cases, the silhouette index is only used with the goal of evaluating the quality of the obtained clustering. Furthermore, in [18], this index is used as an indicator to choose the best clustering technique. Unlike the previous references, in this paper, the silhouette index is used to correct the mapping of a given input.

III. Problem Formulation

Diagnosis is one of the most critical functions in a self-healing network. It is, therefore, essential to ensure that the automatic fault identification is accurate and reliable, so that experts do not have to check every diagnosis provided by the system. Consequently, the automatic diagnosis system must emulate the manual process followed by an expert to determine the existing faults.

Through this manual procedure, experts analyze the symptoms or measurements that may reveal the cause of the problem. These symptoms could be alarms, key performance indicators (KPIs), configuration parameters, etc. Thus, depending on which symptoms are degraded and their level of deterioration, experts can differentiate the fault of a cell. However, this raises difficulties that experts must face, i.e., difficulties that are also present when the diagnosis system is designed. First, the system must be able to operate with a wide range of symptoms. Furthermore, each symptom has different features, e.g., it can be either discrete or continuous, its range of variation can be limited or not, etc. Finally, the automatic diagnosis system, similar to experts, must "know" the effects that each fault causes on the symptoms to perform identification from a plurality of faults.

In addition, automating the diagnosis process implies that the diagnosis system has to learn how the faults behave. A possible approach could be to extract the information from stored cases that have been solved satisfactorily and whose fault is known (i.e., labeled cases). This data set will allow obtaining an automatic system through supervised learning. Nevertheless, since experts do not tend to collect the values of the KPI along with a standard label associated to the cases that they resolve, the available historical records are characterized by being scarce. In particular, they do not have a high variety of faults, and for each specific fault, there is not a high number of labeled cases. As a result, the historical data obtained from a real network is not sufficiently rich to build a diagnosis system with supervised techniques. Although this valuable information could be obtained from expert knowledge, this would give rise to a system that would highly depend on the notions provided by the experts and how this knowledge is translated into the system. Furthermore, each network may have different faults, and not all of them may be known by the experts. Consequently, the unsupervised technique proposed in this paper is the best solution for the design of the automatic diagnosis system, since it avoids the use of labeled cases for its construction.

Unsupervised methods allow building systems from a data set taken directly from the real network, without including any information about the fault cause in question. Moreover, the data set used to design the system contains symptoms from both healthy and faulty cells because these two states cannot be distinguished, i.e., the data set is unlabeled. Therefore, one additional aspect to consider, arising from the use of unsupervised techniques, is that the obtained system must be able to identify not only the cause of the problems but whether a cell has a problem or not as well. As a result, the automatic system simultaneously performs the diagnosis and the detection functions of the self-healing procedure. Although the data set is previously filtered by the detection phase, there will be some samples of healthy cells (due to nonidealities of the detection phase) that arrive to the diagnosis phase. This leads to a diagnosis system that is trained with both healthy and problematic cases. Thus, those healthy cases should also be identified.

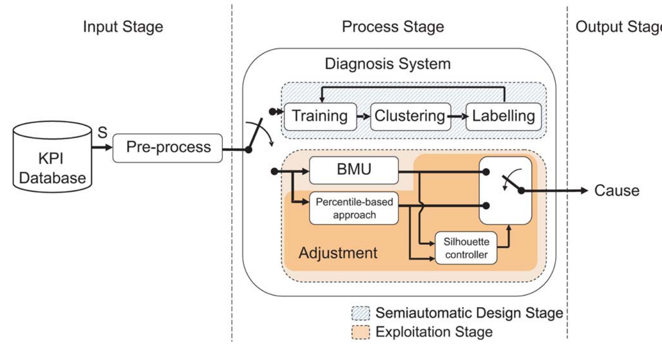

Fig. 1. Automatic root cause analysis.

IV. Proposed Diagnosis System

The root cause analysis procedure can be generally formulated through the scheme shown in Fig. 1, where $S = [KPI_1, KPI_2, \ldots, KPI_M] \in \mathbb{R}^M$ is the input vector that represents the state of the cell by M KPIs, and c is the output of the system whose value belongs to the set $C = \{FC_1, \ldots, FC_L, N\}$. Therefore, the output c determines whether the cell presents a normal behavior (N) or not, indicating, in the latter case, its fault cause (FC). First, in the input stage, the specific KPIs are selected and preprocessed. Then, the preprocessed input vector flows toward the diagnosis system. In particular, the diagnosis system proposed in this paper follows the scheme shown in Fig. 1, and its core is based on SOM. As it can be observed, this system can be executed in two different modes, i.e., one corresponding to the design phase (the cross-hatched area in Fig. 1) and another corresponding to the exploitation phase (the hatched area in Fig. 1). In the first phase, the diagnosis system is built, whereas in the second phase, the system is used to diagnose the fault cause. Each of these three stages is described in detail below.

A. Input Stage

The input data vector (S) consists of all relevant KPIs of the cell under study. These KPIs can be estimated with different time aggregation levels (hourly, daily, weekly, monthly, etc.) according to the required granularity (level of detail) of the diagnosis process. Two types of input data can be distinguished depending on whether the data are used in the design stage or in the exploitation stage.

- Training data: To build the proposed system, a training data set with as many cases as possible is required. The resulting system will depend greatly on the training data set; thus, the training data and the data to be used in the exploitation stage must have the same features, e.g., the same KPIs, time and layer (per cell, per base station, etc.)

aggregation levels, etc. Since no labels are used to obtain the training data set, there will be data describing normal behavior and cells with abnormal behavior due to faults. It is important to highlight that since the data set will include data both from normal cells and faulty cells, one of the classes obtained after the clustering phase will be the normal profile, whereas the rest of classes will correspond to faults.

- Analytical data: During the exploitation stage, the input data are the specific cell's state taken directly from the cell under study.

The input of the diagnosis system must be preprocessed to suit the technical requirements of the system. In particular, for the proposed system, the input data must be quantitative, i.e., they can be expressed in terms of numbers (e.g., power and throughput). In particular, performance indicators of mobile networks are characterized by being numerical variables. As a consequence, this system is appropriate for automating the troubleshooting process in a mobile network by working directly with KPIs, avoiding both the discretization of the variables and the definition of the thresholds by experts.

However, given that the proposed system is based on the Euclidean distance, the raw data taken from the network must be normalized. This ensures that their dynamic ranges are similar, and thus, there are no high values that dominate the training. In this system, the normalization process is performed by using the following methods.

- *Range normalization*: This method transforms the dynamic range of a particular metric (KPI_i). The objective is to ensure that all input variables range within the desired interval. In particular, this method is applied only to those KPI whose values are not within the interval [0, 1]. This normalization is given by the following equation on the basis of the minimum and maximum values of the KPI i in the training data set $(\widetilde{KPI_i})$:

$$\widehat{KPI}_i = \frac{KPI_i - \min(\widetilde{KPI_i})}{\max(\widetilde{KPI_i}) - \min(\widetilde{KPI_i})}. \quad (1)$$

- *Z-score normalization*: This technique modifies the input variable to achieve zero mean and unit standard deviation. It is carried out, taking into account the mean and the standard deviation (std) of $\widetilde{KPI_i}$, through the following linear operation:

$$\widehat{KPI}_i = \frac{KPI_i - \text{mean}(\widetilde{KPI_i})}{\text{std}(\widetilde{KPI_i})}. \quad (2)$$

In this case, this normalization is applied to all input variables to guarantee that all KPIs have unit variance.

B. Semiautomatic Design Stage

Before the proposed diagnosis system can be used, it must be designed through an iterative process (see Fig. 1) with the goal of obtaining an accurate system. The proposed design method consists of different unsupervised techniques with the SOM as the key algorithm, along with the validation of the experts who determine if the obtained diagnosis system makes sense. Thus, this semiautomatic design is carried out through three different phases: unsupervised SOM training, unsupervised clustering, and labeling by experts.

1) Unsupervised SOM Training: An SOM is a type of unsupervised neural network capable of acquiring knowledge and learning from a set of unlabeled data. Therefore, in this paper, SOM is used as the centerpiece to classify the cell's state according to the behavior of its symptoms, subsequently identifying the fault cause. The great advantage of SOM is its capacity for processing high-dimensional data and reducing it to a lower dimension (e.g., two), which enormously facilitates the interpretation and understanding of the final diagnosis. Furthermore, this system does not require discrete data. This enables working directly with raw data, with no discretization methods that cause loss of information.

In particular, SOM consists of elements (artificial neurons) that are organized in a 2-D grid. Each of those neurons has a specific weight vector $(W = [W_{KPI_1}, W_{KPI_2}, \ldots, W_{KPI_M}] \in \mathbb{R}^M)$ whose dimension M is determined by the number of KPIs in the input vector. First, all weight vectors are initialized, for example, through the linear initialization method described in [10]. Via this method, the weight vectors are initialized in an orderly fashion along 2-D subspace spanned by the two principal eigenvectors of the training data set [19]. The advantage of the linear initialization is that the convergence of the algorithm is much faster than if the random method is used. Then, these weight vectors are updated through an unsupervised training process to determine the values of the weight vectors that best match the behavior of the input data. Fundamentally, the training process depends on the training data and the neighborhood function (h_{ij}) that links the neurons i and j, and it consists of identifying the winner neuron or the best matching unit (BMU) and subsequently updating both its weight vector and the weight vector of all its neighboring neurons.

In this paper, the SOM training is done in two phases. The first phase is the rough phase, which aims to order the neurons, whereas the second phase is the fine phase that intends to achieve the convergence, as follows.

- Rough phase: First, the training in the rough-tuning phase is carried out by the Batch algorithm [19] (summarized in Algorithm 1) due to its rapid convergence [20], particularly in the cases where the linear initialization method is used. The Batch algorithm is an iterative method that is characterized by modifying all the weight vectors after the entire data set has been presented to the SOM. At the beginning of each iteration (t), for each normalized input $(\hat{S}_i)$, the winner neuron, namely, the BMU (n_i^t), is searched in relation to the minimum Euclidean distance (d). Afterward, the weight vectors (W^t) of all neurons are updated, taking into account the neighborhood bubble function $(h^t_{n_i^t j})$ [21] and its radius(σ^t). In particular, that function determines the neurons around n_i^t that are considered neighbors and, thus, the neurons that must be updated. The rough-tuning phase aims to order the neurons quickly; therefore, it is carried out in a few thousand iterations. Furthermore, at the beginning of the rough phase, the radius of the neighborhood function covers all neurons, and it linearly decreases at each iteration until it only covers two neighboring neurons [22].

Algorithm 1. A Batch SOM: Rough-tuning

1. Calculate the BMUs for all input vectors:

$$n_i^t = \arg\min_j \left\| \hat{S}_i - W_j^t \right\|, \ i \in [1, \ldots, P]$$

where P is the number of input vectors in the training data set.

2. Apply neighborhood function (Bubble):

$$h_{n_i^t j}^t = \begin{cases} 1, & \text{if } d\left(n_i^t, j\right) < \sigma^t \\ 0, & \text{if } d\left(n_i^t, j\right) > \sigma^t \end{cases}$$

where σ^t is the neighborhood radius in the current iteration t:

$$\sigma^t = \sigma^{(t-1)} - \frac{\sigma^0 - \sigma^{T_1}}{T_1}$$

σ^0 and σ^{T_1} are the initial and final neighborhood radius, respectively, and T_1 is the number of iterations.

3. Update the weight factors:

$$W_j^{(t+1)} = \frac{\sum_{i=1}^{P} h_{n_i^t j}^t \times \hat{S}_i}{\sum_{i=1}^{P} h_{n_i^t j}^t} \qquad \forall\, j$$

4. Repeat the above-described process starting from step **1** until t $= T_1$

- Fine phase: In the fine-tuning phase, the sequential algorithm [19] (summarized in Algorithm 2) is used. In each iteration, a random normalized input $(\hat{S}_i)$ is presented, and its winner neuron (n^t) is searched. Thereafter, the weight vectors of the winner neuron (W_n^t) and its neighbors (W_j^t) are updated considering the neighborhood function $(h_{n^t j}^t)$ and the learning rate (α^t). The objective is to slowly modify the weight vectors until the convergence of the system is achieved. This process is repeated during several thousand iterations. Thus, the learning rate must take very small values (e.g., 0.01), and, unlike in the previous phase, the neighboring radio only includes the closest neurons (e.g., one) [22].

Algorithm 2. Sequential SOM: Fine-tuning

1. Calculate the BMUs:

$$n^t = \arg\min_j \left\| \hat{S}^t - W_j^t \right\|$$

2. Neighborhood function: Gaussian

$$h_{n^t j}^t = e^{\frac{-\left\| W_n^t - W_j^t \right\|^2}{2(\sigma^t)^2}}$$

3. Learning rate:

$$\alpha^t = \frac{\alpha^0}{1 + 100\frac{t}{T_2}}$$

where α^0 is the initial learning rate, and T_2 is the number of iterations.

4. Update step:

$$W_j^{(t+1)} = W_j^t + \alpha^t h_{n^t j}^t \left[\hat{S}^t - W_j^t \right] \ \forall\, j,$$

$$\sigma^{(t+1)} = \sigma^t - \frac{\sigma^0 - \sigma^{T_2}}{T_2}$$

where σ^0 and σ^{T_2} are the initial and final neighborhood radius, respectively.

5. Repeat the above-described process starting from step **1** until t $= T_2$

After this SOM training, the topology of the neural network will be as the spatial distribution of the training data set.

2) Unsupervised Clustering: In this phase, all ordered neurons of the SOM system are clustered in as many groups or classes as possible using an unsupervised algorithm. To that end, it must be noted that the activated neuron (i.e., the BMU) for a specific input or a cell's state is the closest based on the Euclidean distance (d). Thus, a similar cell's states activate nearby neurons or even the same neuron. This indicates that the same cause $(c_i \in C)$ can be represented by several neurons with slight differences between them; hence, they must be grouped together into the same class $(g_i \in G)$. Therefore, once the SOM has been trained, the next step groups the neurons representing data with similar behavior into the same cluster. The aim is to divide the SOM map into as many clusters as causes can be distinguished.

It is highlighted that the exact number of clusters in the data set is unknown because this system works with unlabeled data; hence, there is no evidence to reliably determine it in advance. As a result, two design considerations are established to automatically determine whether the obtained classification can be accepted or not. First, the identification of all existing clusters in the data set is not necessary because the objective is to identify the most frequent causes, that is, the most relevant causes to recognize which are the most problematic faults and automate the diagnosis of the most repeated problems. Consequently, the total number of clusters that are identified by the system should be limited. In particular, the system should identify a minimum of two clusters (the normal state and a faulty state) and a maximum TC configured by the designer (e.g., $TC = 10$). Second, all the identified clusters are valid only if they present different statistical behaviors. Taking these points into consideration, we propose to cluster the neurons in an unsupervised manner through the following algorithm.

- Let $G = \{g_1, \ldots, g_J\}$ be the set of classes just after the training; in particular, this method starts with a class for each neuron.
- Then, SOM is clustered into different number of groups (from 2 to TC) using Ward's hierarchical clustering

algorithm [23]. To that end, the Euclidean distance between each pair of classes $(d(g_j, g_k))$ is calculated. Then, the Ward algorithm iteratively merges the closest two classes into a new class based on the minimum distance. After each union, the distances between each current pairs of clusters are updated following the Lance–Williams recurrence formula [23].

- Each classification is evaluated through the DB index [24], which determines how well is each clustering, and the clustering with the minimum index (i.e., the best clustering according to the DB metric) is selected.
- Finally, it is verified that no pair of clusters have a similar statistical behavior through the KS test [7]. The null hypothesis to be checked by this test is that the observed values of a KPI of two clusters present the same distribution. Therefore, the KS test is applied to each KPI for each pair of cluster using as observed values the values of the training data set that have been assigned to those clusters. The p-value obtained for each KPI and pair of cluster determines the probability of having values consistent with the null hypothesis. In particular, there must not be any pair of clusters whose KPIs present the same behavior. The lower the p-value, the more inconsistent is the data with the null hypothesis; hence, if the p-value is too small, the null hypothesis can be rejected. Taking this into account, we consider that two clusters are statistically different if the p-value obtained for all their KPIs are below a predefined threshold called the significance level (e.g., $SL = 0.1\%$). If there is at least one pair of cluster that is statistically similar, the number of clusters is reduced, and the clustering obtained with Ward's hierarchical algorithm for the new number of clusters is chosen. This process is repeated until all the identified classes are statistically different. As a result, the final set of classes is $G = \{g_1, \ldots, g_L, g_N\}$ (i.e., L fault causes along with the normal case (N)).

To explain this, let us assume a cell's state consisting of two KPIs: retainability, which indicates the rate of connections that have finished properly to the total number of connections that have occurred in the cell; and handover success rate (HOSR), which determines the rate of handover performed successfully to the total number of handovers. Let us assume that the data set, for example, is composed of a cell's states belonging to three different causes, but this detail is unknown so, in the clustering phase, the DB index has indicated that the best number of classes is four (see Fig. 2). However, the KS test determines that clusters 3 and 4 present similar statistical behavior. This means that they represent the same cause, since both cases have the same cluster-symptom relation. Thus, the number of cluster is automatically reduced, and the KS test is applied to the new clustering.

3) Labeling by Expert: At this stage, the obtained clusters need to be labeled with the identified causes. First, the unsupervised phases of the design (training and clustering) must be verified because there is no information to evaluate the goodness of the solution. A simple way of doing that is to verify if the solution is right by a simple visual inspection of the neurons belonging to each group. To illustrate this, Fig. 3(a) shows an

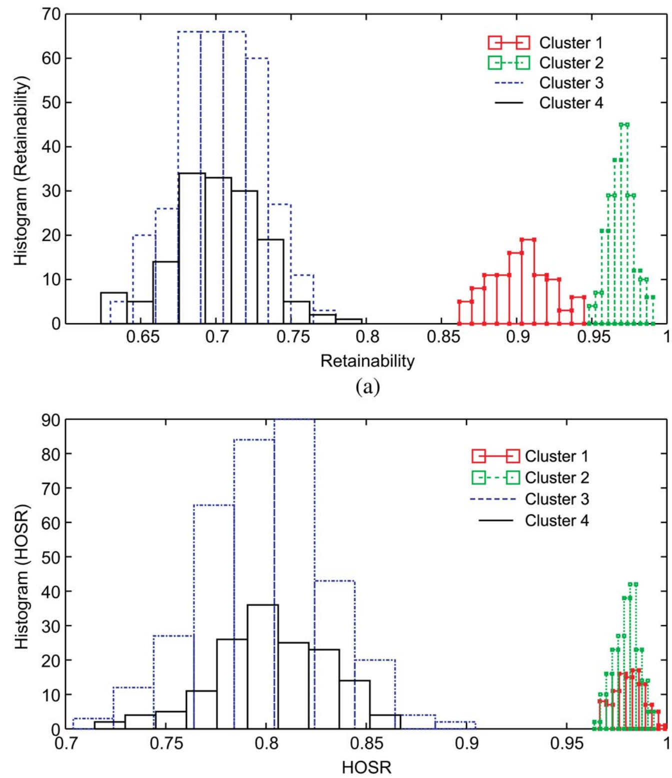

Fig. 2. Histograms of (a) retainability and (b) HOSR given each cluster when the neurons have been grouped into four clusters.

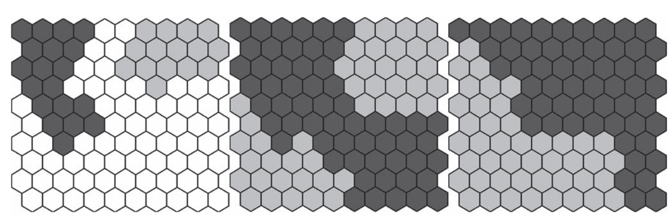

Fig. 3. Neurons of the diagnosis system during (a) the clustering phase and the resulting classification (b) with fragmented clusters and (c) without fragmentation.

example of an SOM of 10×10 neurons during the clustering phase where the neurons are being grouped into two different classes. In each iteration, a new neuron (represented in white) is selected and joined to the closest group. At the end of the clustering, all neurons belong to one of the created groups. However, if the resulting classification provides a fragmented cluster (see Fig. 3(b)), then the training and clustering phase must be repeated until clusters are composed of adjoining neurons as in Fig. 3(c). This can be done by changing the parameters of the training procedure, such as the training length, the initial or final neighborhood radius, and the learning rate of the sequential algorithm.

Furthermore, since there is a nondeterministic relation between KPIs and causes [3], it is necessary to analyze their statistical relations. To do this, the following process is proposed.

- For each cell's states contained in the training data set, the associated cluster must be identified. In particular, the cluster (g_j) associated with a particular normalized cell's state $(\hat{S}_i)$ is considered to be that which includes the neuron (BMU) activated for that state $\hat{S}_i$, i.e.,

$$\hat{S}_i \in g_j \leftrightarrow BMU(\hat{S}_i) \in g_j. \quad (3)$$

- Once all of the cell's states in the data set have been included in a given cluster, the conditional probability density functions (pdfs) of each KPI given each cluster $(f(KPI_i|g_j))$ are estimated. As the distribution followed by the KPI_i is unknown, a nonparametric technique must be used. Among them, the most commonly used to define the pdf is the histogram or the kernel smoothing function [25].
- The estimated pdfs for each cluster are studied to examine the statistical behavior of each KPI and, as a result, determine the cluster-symptom relation. This statistical information also helps verify whether the clustering is correct or not. That is, it allows experts to detect if a cluster is associated with more than one cause.
- Finally, taking into account the cluster-symptom relations, the experts should identify the cause associated with each cluster based on their knowledge and, as a consequence, provide a suitable label to each cluster. As previously stated, one of the clusters will correspond to the normal behavior of the cells, and it will be labeled with N; on the other hand, the remaining cases will have a descriptive label related with the possible fault cause (FC_i). Therefore, the process of labeling maps the clusters to a specific cause, i.e.,

$$G = \{g_1, \ldots, g_L, g_N\} \rightarrow C = \{FC_1, \ldots, FC_L, N\}. \quad (4)$$

C. *Exploitation Stage*

Once the system has been designed, it can be used to automatically perform the diagnosis in the exploitation phase. This diagnosis process can be periodically performed to determine whether the identified fault is sporadic, continuous, or periodic and to track the evolution of the fault over time when either compensation or recovery tasks are carried out.

The proposed diagnosis process is summarized in Algorithm 3. Such a process must be as accurate and reliable as possible. To achieve this, first, for a specific cell's state (S_i), the winner neuron (BMU) is determined on the basis of the minimum Euclidean distance, and thus, the diagnosis $(Diagnosis_{BMU})$ is the cause (c_j) related to that neuron. However, if the activated neuron is at the border between two or more causes, the likelihood of an erroneous diagnosis is higher because the cell's state is close to different faults' behaviors. Thus, we propose further processing to guarantee a high successful diagnosis rate. In particular, the adjustment proposed in this paper is carried out using a percentile-based approach and a silhouette controller (see Fig. 1).

- Percentile-based approach: This method is only used when the BMU is a border neuron among different causes. In this situation, the diagnosis should be the cause located at the borderline that has, in general, the most similar behavior. In view of the above, a different approach to identify the most similar cause is proposed. For each border cause, the Xth percentile of the distances between the cell's state and all neurons in that cluster are estimated. Then, the cause selected by the percentile-based approach $(Diagnosis_P)$ is that which has the minimum Xth percentile of all its distances. Compared with the BMU where the considered distance is only to one single neuron (i.e., the closest), the proposed approach considers the distance to all neurons in the cluster.
- Silhouette controller: Once the diagnoses have been determined by the BMU and percentile-based approaches, one of the two diagnoses must be selected. For this purpose, a controller based on the silhouette index [8] is proposed. First, if the diagnosis provided by the percentile-based approach matches that given by the BMU, it can be concluded that the diagnosis is right, and as a result, the selected cause corresponds to the $Diagnosis_{BMU}$. Nevertheless, when both diagnoses are different, a comparative evaluation is required so that it is possible to discern which one is better. Therefore, the average silhouette is used to evaluate the quality of each diagnosis and determine whether the input fits well with the selected clusters or not. Then, the silhouette index for each neuron and the input vector are estimated through the equation of step 3.3.1 in Algorithm 3, shown below. Particularly, this process is carried out twice, i.e., once for the mapping made with the BMU and once for the mapping of the percentile-based approach. For each diagnosis, the average silhouette is calculated. This measure shows how well the input has been categorized in a cause. In particular, the higher the average silhouette is, the better the classification is. Consequently, the final decision is the diagnosis whose average silhouette is greater.

Algorithm 3. Automatic diagnosis in the exploitation phase:

---

1. Calculate BMU for the state S

$$BMU(\hat{S}) = \arg\min_j \left\| \hat{S} - W_j^T \right\|.$$

2. **if** $BMU(\hat{S})$ is not at the border

$$Diagnosis(\hat{S}) = Diagnosis_{BMU} = c_j \in C \leftrightarrow BMU(\hat{S}) \in c_j.$$

3. **else**:

3.1. Calculate Euclidean distance to each neighboring cause:

$$D_{c_{neigh}} = \left\{ d\left(\hat{S}, W_j^T\right) : j \epsilon\, c_{neigh} \right\} \forall\, neigh.$$

3.2. Determine the most similar neighboring cause:

$$Diagnosis_P = c_{neigh} \in C \leftrightarrow X^{th} \; percentile\ of\ D_{c_{neigh}}\ is\ minimum.$$

3.3. **if** $Diagnosis_{BMU}$ and $Diagnosis_P$ are different.

3.3.1 Calculate the silhouette associated to both diagnoses (BMU and P): using all the neurons (J) and the input

$$Silhouette_x(i) = \frac{b(i) - a(i)}{\max\left(b(i), a(i)\right)},\ i \in [1, \ldots, J+1]$$

where x determines whether the silhouette is related to $Diagnosis_{BMU}$ or $Diagnosis_P$, $a(i)$ is the average Euclidean distance between i and all the other components of its cluster, and $b(i)$ is the minimum average Euclidean distance between i and all the components of the nearest neighboring cluster different to the cluster of i.

3.3.2 Calculate the average silhouette for both diagnoses (BMU and P)

$$\overline{Silhouette_x} = \frac{1}{J+1} \sum_{i=1}^{J+1} Silhouette_x(i), \; x = \{BMU, P\}.$$

3.3.3 Choose the final diagnosis:

$$Diagnosis(S) = \begin{cases} Diagnosis_{BMU} & \text{if } \overline{Silhouette_{BMU}} \geq \overline{Silhouette_P} \\ Diagnosis_P & \text{if } \overline{Silhouette_P} > \overline{Silhouette_{BMU}}. \end{cases}$$

It should be noted that during the exploitation stage of the system, only a specific diagnosis (i.e., one cause) is provided, corresponding to the diagnosis that best matches the values of the KPIs measured in the faulty cell. Therefore, when a cell presents deterioration due to multiple causes, the diagnosis provided by the system depends on the effect of those multiple causes on the KPIs. On the one hand, a cell deteriorated due to multiple causes can present the symptoms of the most dominant fault. In this case, only the most dominant cause is identified by this system. Once this fault is solved, then the next dominant fault can be identified. On the other hand, if a specific combination of multiple causes produces its particular symptoms, this behavior could lead to a specific cluster of the diagnosis system during the design stage (provided that there are enough cases with this behavior in the training data set). Therefore, in this situation, this combination of multiple causes will probably be diagnosed with the label assigned by the expert.

V. Experiment Results

Herewith, the assessment of the proposed diagnosis system is presented. First, the characteristics of the simulated data are described. Afterward, the proposed diagnosis system is built for the simulated LTE network, illustrating the construction process. Then, with a labeled data set, the obtained system is evaluated and compared with reference mechanisms. Furthermore, the complexity of the proposed system is discussed. Finally, a demonstration of the diagnosis system in a live network is presented.

A. Simulated Data Set

This section briefly presents the features of the used data set, which are available at the site provided in [26]. This data set consists of two different collections: the training set, which will be used to design the proposed diagnosis system, and thus, it is used without labels; and the validation set, which will be used to evaluate the system. In particular, the data set has been generated by a dynamic LTE system-level simulator implemented in MATLAB [27], whose 57 macrocells are evenly distributed across the entire scenario, forming a hexagonal grid. Table I describes the principal parameters of the simulator.

TABLE I
SIMULATION PARAMETERS

Parameter	Configuration
Cellular layout	Hexagonal grid, 57 cells, cell radius 0.5 km
Transmission direction	Downlink
Carrier frequency	2.0 GHz
System bandwidth	1.4 MHz (6 physical resource blocks)
Frequency reuse	1
Propagation model	Okumura-Hata with wrap-around, Log-normal slow fading, σ_{sf} = 8 dB and correlation distance=50m
Channel model	Multipath fading, ETU model
Mobility model	Random direction, 3 km/h
Service model	Full Buffer, poisson traffic arrival
Base station model	Tri-sectorized antenna, SISO, Donwtilt=9 ° $P_{TX_{max}}$ =43 dBm, Azimuth beamwidth (AB)=70 ° Elevation beamwidth (EB)=10 °
Scheduler	Time domain: Round-Robin, Frequency domain: Best Channel
Power control	Equal transmit power per physical resource blocks
Link Adaptation	Fast, CQI based, perfect estimation
Handover	Triggering event = A3, HOM = 3 dB, Measurement type = RSRP
Radio Link Failure	SINR < −6.9 dB for 500 ms
Traffic distribution	Evenly distributed in space
Time resolution	100 TTI (100 ms)
Epoch & KPI time	100 s

To carry out the root cause analysis of this scenario, the chosen indicators are statistics at the cell level related both to radio environment quality and the efficiency of the offered service. In particular, the used KPIs are those proposed in [28].

- retainability, which is the ratio of connections that have finished successfully to the total number of finished connections;
- HOSR, which is calculated as the ratio of the number of successful handovers to the total;
- the 95th percentile of the reference signal received power ($RSRP_{95}$) measured by all the users connected to the cell. The RSRP is defined in [29] as the average power received from the serving cell over the resource elements that carry the cell-specific reference signals (RS) within the considered measurement frequency bandwidth;
- the fifth percentile of the reference signal received quality ($RSRQ_5$), which is the ratio of the RSRP multiplied by the total number of resource blocks to the total received power within the measurement bandwidth;
- the 95th percentile of the signal-to-interference-plus-noise ratio ($SINR_{95}$). In particular, the SINR of the users is the ratio of the desired power received to the total power of noise and interference;
- the average throughput of all users in a cell ($AvThroughput$), where the user throughput (T_u) is calculated on the basis of their SINR through the following equation [30]:

$$T_u = (1 - BLER(SINR_u)) \cdot \frac{D_u}{TTI} \tag{5}$$

where $BLER$ is the block error probability that depends on the SINR of user u, D_u is its data block payload in bits, and TTI is the transmission time interval;

- the 95th percentile of the distance between the base station and each user ($Distance_{95}$);
- therefore, the input vector of the system consists of the previous KPIs; formally, it can be expressed as

$$S = [Retainability,\ HOSR,\ RSRP_{95},\ RSRQ_5,\ SINR_{95},\ AvThroughput,\ Distance_{95}]. \quad (6)$$

Considering these KPIs, all normal cells (i.e., cells without problems) have been configured to achieve good cell performance in terms of retainability and HOSR (which must be above 0.98 and 0.95, respectively). The KPI time period in this simulation, i.e., the time interval in which the KPIs are estimated, corresponds to a simulation loop that is sufficiently long to provide reliable statistics (in this study, it is composed of 18 000 simulation steps). In each simulation loop, all cells present the normal configuration, except a set of cells that present one of the simulated faults.

In particular, six fault causes (presented in [28]) have been simulated to deteriorate some randomly chosen cells. The first cause consists of an excessive uptilt (EU) of the antenna, which causes overshooting due to the excessive increment of the coverage area. The second fault is caused by an excessive downtilt (ED) of the antenna due to a wrong parameter configuration. This causes a reduction in the coverage area from planned, focusing the transmission power of the cell near the base station. The third fault cause is modeled by excessively reducing the transmission power of the antenna [it is called excessive reduction of cell power (ERP)]. This can happen in a real network due to wiring problems or a wrong parameter configuration. The next fault cause consists of creating a coverage hole (CH) within the coverage area of a cell by increasing the attenuation suffered in a small part. This models the shadowing caused by obstacles in the environment (such as buildings or hills). Another simulated fault cause is a too late handover (TLHO) problem. If the handover margin (HOM) parameter is wrongly configured, the handover process is not correctly performed. In particular, the handover is performed too late when the HOM is too high; hence, the condition to perform the handover is very restrictive. The last fault cause is an intersystem interference (II) problem that may happen due to external systems such as TV, radars, or even other cellular systems. This is simulated, adding an extra antenna in the scenario within the service area of a cell. The configuration used for generating each fault cause is presented in Table II.

Each case represents the state of a cell, i.e., it is a vector with the value of its KPI estimated over a KPI time period. Therefore, in each simulation loop, a case for each cell of the scenario is obtained. Since, in a live network, the frequency in which faults happen is not deterministic, the number of cases stored for each fault has been randomly selected, also taking into account that the frequency of each fault is different. The actual number of cases that compose the data sets is presented in Table II. It is highlighted that normal cells are more common than faulty cells in a real network, and thus, the total number of normal cases is much greater than the number of fault cases in both simulated data sets. This allows determining if the system can identify both the most and the less prevalent problems within the data set. Another important detail is that the size of the training data set is much lower than the size of the validation data set. This is to ensure that the system can be properly designed with little data and that there are sufficient data available to estimate the overall total error.

TABLE II
FAULT CAUSE DESCRIPTION

Fault Cause	Configuration	Number of Cases	
		Training	Validation
EU	Downtilt=[0,1] °	32	212
ED	Downtilt=[16,15,14] °	28	212
ERP	ΔP_{TX} =[7,8,9,10] dB	28	208
CH	Δ_{hole} =[49,50,52,53] dBm	14	103
TLHO	HOM=[6,7,8] dBm	34	204
II	$P_{TX_{max}}$ =33 dBm Downtilt=15 ° AB=[30, 60] ° EB=10 °	15	106
No fault	Normal	399	2964

B. Experimental Design

Here, the proposed diagnosis system is applied to the simulated LTE network to diagnose the cells and find abnormal behaviors due to faults. In particular, the system has been implemented in MATLAB using its Statistic Toolbox and the SOM Toolbox [31]. The diagnosis system has been designed as proposed in Section IV, using only the training database (described in Section V-A and without considering any label for each case).

First, the system has been trained with the configuration parameters presented in Table III. The configuration of the training data set has been done in accordance with the theoretical concepts. That is, the rough phase has been carried out in a thousand iterations, whereas the fine phase has been performed in 20 000 iterations, but using a smaller neighborhood radius.

After that, the neural network has been clustered in an unsupervised manner because there is no information available about the specific fault cause associated with each group. To that end, the two configuration parameters of the unsupervised clustering algorithm should be set. In this paper, the maximum number of cluster identifiable by the system (TC) has been set at 10. Therefore, the final number of cluster is limited to a value between 2 and 10. This ensures that the system analyzes a wide enough range of possible clustering to find the top ten clusters. The significance level (SL) for the KS test has been set to a small value (i.e., 0.1%) to ensure that the statistical behavior of the identified clusters is sufficiently different. During this process, the DB index is calculated to each of the clustering performed with Ward's hierarchical method. In Table IV, the DB index for each possible clustering is presented. As shown, the clustering consisting of eight clusters presents the smallest DB index; this means that it is the best clustering according to the DB index. Then, the KS test is applied between each pair of clusters to compare the distribution of their KPIs. To

TABLE III
CONFIGURATION PARAMETERS USED TO TRAIN THE SYSTEM

Parameters	Simulated scenario		Real scenario	
	Rough-tuning	Fine-tuning	Rough-tuning	Fine-tuning
Training algorithm	Batch	Sequential	Batch	Sequential
Training lenght, T	1000	20000	1000	2000
Neighbourhood function	Bubble	Gaussian	Bubble	Gaussian
Initial neighbourhood radius	10	1	10	1
Final neighbourhood radius	2	0	1	0
Initial learning rate	-	0.01	-	0.001

TABLE IV
DB INDEX

Number of Clusters								
2	3	4	5	6	7	8	9	10
1.43	1.25	1.06	0.98	0.84	0.80	0.7	0.83	0.87

TABLE V
P-VALUE OF THE KS TEST BETWEEN TWO CLUSTERS

KPI						
Ret	HOSR	$RSRP_{95}$	$RSRQ_5$	$SINR_{95}$	AvTh	$Dist_{95}$
0.0026	0.0026	0.3788	0.0045	0.8552	0.2853	0.2095

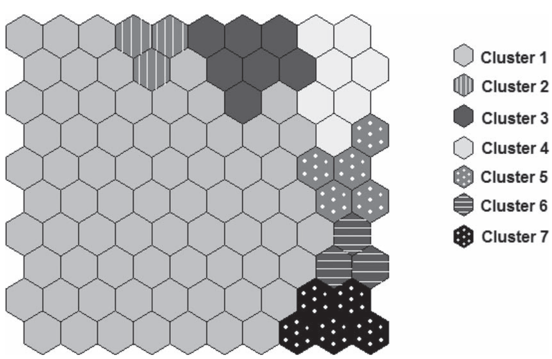

Fig. 4. Proposed diagnosis system clustered in seven groups by the unsupervised clustering method.

that end, the cases of the training data set that are assigned to each cluster are used. As an example, the p-value obtained by applying the KS test to the KPIs of a specific pair of clusters is shown in Table V. It can be seen that the p-value obtained for all KPIs is greater than the predefined significance level (i.e., 0.001). This means that the distribution of the KPI of both clusters is very similar. Therefore, the KS test reveals that these clusters present similar statistical behaviors. Consequently, the number of cluster is reduced, and the KS test is repeated. In this case, the KS test determines that there is no pair of clusters with the same statistical behavior; hence, the automatic unsupervised clustering concludes and determines that the best number of clusters is seven. As a result, neurons are grouped as shown in Fig. 4. With these two evaluations, the proposed clustering ensures that the chosen classification is the best according to the DB index, or at least, it does not include any cluster with similar statistical behavior to another according to the KS test.

Afterward, in the labeling phase, the statistical behavior of each cluster has been analyzed in detail to reasonably identify their corresponding cause. This requires knowledge and understanding of when a specific KPI is degraded, but this information usually depends on the network features and context factors, and it is not always well defined. Therefore, first of all, the pdfs of each KPI estimated by the kernel smoothing function are analyzed to determine whether each KPI is deteriorated for each cause (see Fig. 5). Due to the fact that we are using the kernel smoothing function, the pdfs are estimated through the sum of Gaussian functions centered at the data. Therefore, the estimated pdf is smoother than the histogram causing the estimated pdf to cover values out of range [see Fig. 5(a) and (b)]. It is important to highlight that this is part of the estimation error; however, this particular error does not affect the proposed analysis, since the estimated pdfs are only used to determine the overall statistical behavior of the clusters by visual inspection.

The objective of this examination is to get a rough idea about the most probable values of each KPI, depending on the cluster and, thus, determine, for a specific cluster, if the majority of the values of a KPI is considered damaged or not. As a result of this detailed study, the overall behavior of each cluster has been assessed to identify the associated fault cause and assign it a label (see Fig. 5 and Table VI).

- Cluster 1: This cluster represents cells with normal behavior given that none of the indicators are deteriorated.
- Cluster 2: Comparing the pdfs of the second cluster with those of the first cluster, it is possible to identify that this cluster represents cells whose KPIs have a normal behavior but with the difference that their retainability is more likely to be low (approximately below 0.98). Therefore, the high number of dropped calls together with the fact that no other KPIs are deteriorated means that the fault cause is a CH within the service area. This results from the fact that the CH affects only a particular group of users located in this specific area, so that the aggregated values of the other analyzed KPIs do not present any deterioration.
- Cluster 3: After analyzing the pdfs, it is possible to conclude that the most deteriorated KPIs of this cluster are retainability, $HOSR$, and $RSRQ_5$ because they are approximately below 0.98, 0.9, and -18.5 dB, respectively. Bad $HOSR$ determines that this cell has mobility problems, whereas low $RSRQ_5$ shows that the number of users with bad quality has increased. This reveals that the cell is retaining users with bad quality instead of performing the handover process. As a result, these kinds of cells presents mobility problems due to the fact that handovers are carried out too late.
- Cluster 4: All the KPIs of this cluster, except average throughput, are likely to be low. That is, the overall performance of those cells is degraded. The number of drops has increased, and the maximum served distance ($Distance_{95}$) has been reduced (approximately below -75 dBm and

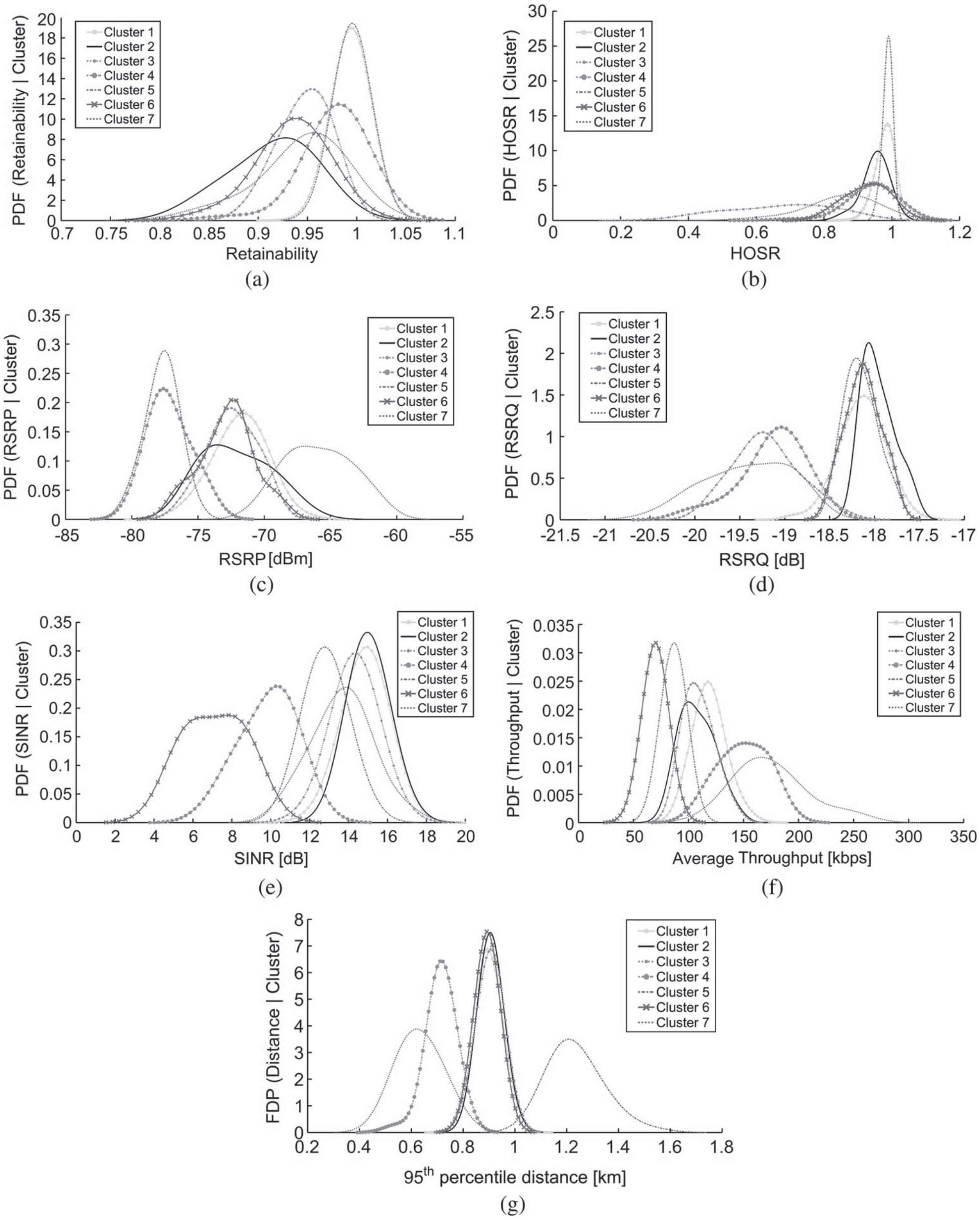

Fig. 5. Estimated pdf of (a) retainability, (b) HOSR, (c) RSRP 95 pctl, (d) RSRQ 5 pctl, (e) SINR 95 pctl, (f) average throughput, and (g) distance 95 pctl given each fault cause.

TABLE VI
LABEL ASSIGNED TO EACH CLUSTER

Clusters	Label
Cluster 1	Normal
Cluster 2	Coverage Hole
Cluster 3	Too late HO
Cluster 4	Excessive reduction of cell power
Cluster 5	Excessive Uptilt
Cluster 6	Inter-system interference
Cluster 7	Excessive downtilt

0.8 km, respectively). This indicates that those cells cannot carry on providing service to the furthest users, which typically have the lowest throughput; hence, the average throughput of the cell has been increased. At the same time, the level of signal received by the best users ($RSRP_{95}$) has been decreased, which reveals that the nearby users have experienced a decrease in their received power. Therefore, this fault is causing deterioration both to nearby and distant users. This behavior corresponds to cells whose transmission power has been considerably reduced.

- Cluster 5: In this cluster, the degraded KPIs are $RSRP_{95}$, $SINR_{95}$ and the average throughput (approximately below −75 dBm, 13 dB, and 100 kb/s, respectively); furthermore, those cells serve more distant users than in normal conditions. Thus, those cells present a service area much greater than necessary. This fault is caused by an EU of their antennas because the coverage area of the cells has been increased, and at the same time, the power received by the nearby users, and thus, their SINRs have been reduced.
- Cluster 6: The pdfs of this cluster show that $SINR_{95}$ is low (approximately below 13 dB), which means that

this KPI has been degraded. As a consequence, in those cells, the quality of service has worsened, causing a lot of drops and a decrease in the average throughput (which is approximately below 100 kb/s). Thus, the performance of those cells is deteriorated due to a high level of II.

- Cluster 7: The last cluster matches a problem caused by an ED of the antenna, where retainability, $RSRQ_5$, and $Distance_{95}$ are low (approximately below 0.98, −18.5 dB, and 0.8 km, respectively), whereas the level of signal ($RSRP_{95}$) is even better than in a normal situation (approximately above −68 dBm). This results from the fact that an antenna with high downtilt focuses its radiation around the base station, causing a reduction in the serving area and the unintended disconnection of the distant users while the signal level of the nearby users improves.

C. Performance Evaluation

Here, the diagnosis system is assessed and compared with reference mechanisms to show its effectiveness. It should be pointed out that there are no previous unsupervised systems proposed in the literature for diagnosis in wireless networks. Among the available supervised solutions in the literature, two reference systems have been chosen. The first reference system is the rule-based system (RBS) [32], which can be considered a baseline scheme in the field of diagnosis, given that it is the simplest solution and widely used by network operators. This system uses a set of $IF \dots THEN$ rules to perform a diagnosis based on the values of the KPIs. The second reference system is a Bayesian network classifier (BNC), which was proposed in [5] to diagnose faults in mobile networks. In this paper, it has been designed using the GeNIe modeling environment [33]. Both systems require discretized inputs; thus, the PBD method proposed in [5] is used. The threshold of this method discretizes each KPIs based on the $X\%$ percentile of the training data, where percentage X is defined by the expert (e.g., 5% percentile of the normal value of each KPI is chosen in this analysis). Furthermore, since the two mechanisms are supervised systems, they require labeled data to be built. In particular, both systems have been built using the training data set, similar to the diagnosis system, but for these reference mechanisms, the label of the data has been used.

In light of the above and despite the fact that the proposed diagnosis system is unsupervised, evaluation has been done with the validation data set using the labels of the cases so that it is possible to compare it with the reference solutions. To do that, three different metrics have been used.

- False Positive Rate (FPR): It is the proportion of normal cases diagnosed as a fault cause (N_{FP}) to the total number of normal cases (N_{NC}). This measurement determines the probability of the system to identify a normal case as problematic, i.e.,

$$\text{FPR} = \frac{N_{FP}}{N_{NC}}. \tag{7}$$

- False Negative Rate (FNR): It is the proportion of problematic cases diagnosed as normal (N_{FN}) to the total number of problematic cases (N_{PC}). It represents the incapacity of the system to detect problems in a cell, when that cell actually has a problem, i.e.,

$$\text{FNR} = \frac{N_{FN}}{N_{PC}}. \tag{8}$$

- Diagnosis Error Rate (DER): It is the proportion of problematic cases diagnosed with a fault cause different to the real one (N_E) to the total number of problematic cases (N_{PC}). It indicates the probability of misdiagnosing a problematic cell, i.e.,

$$\text{DER} = \frac{N_E}{N_{PC}}. \tag{9}$$

Based on these measures, the total error (E_{total}) of the system can be estimated through the following expression:

$$E_{total} = P_N * \text{FPR} + P_{PC} * (\text{FNR} + \text{DER}) \tag{10}$$

where P_N and P_{PC} represent the prevalence of normal and problematic cases, respectively, over the total validation data set. In particular, their values are $P_N = 73.93\%$ and $P_{PC} = 26.07\%$ in the validation data set.

TABLE VII
TEST EVALUATION

System	FPR [%]	FNR [%]	DER [%]	E_{total} [%]
Proposed system	0.24	1.24	1.05	0.77
RBS	37.42	0	16.36	31.93
BNC	10.9	6.2	10	12.28

Table VII shows the evaluation error estimated for each system. Since the simulations to obtain the artificial data set presented in Table II are very time consuming, there is only one validation data set available. As a result, those total errors are derived from only this validation data set, and therefore, it is not an averaged value. However, since the number of cases in the validation data set is very high, the obtained error is a valid average figure. It should be noted that there is only one validation data set available, and therefore, the errors have been calculated from this. From the results, it can be concluded that both RBS and BNC present a higher error rate than the proposed diagnosis system, although they are supervised techniques. This results from the fact that the two reference mechanisms discretize their inputs causing a great loss of information, whereas the proposed system works directly with the continuous value of the KPIs. Therefore, the performance of RBS and BNC is highly dependent on the discretization process. The conclusion to be drawn from the above is that by processing the inputs with higher resolution, the proposed diagnosis system is capable of performing successful diagnoses, even without having considered the labels in the design phase. As it can be seen, the overall error is very low ($E_{total} = 0.77\%$), particularly for an unsupervised system. This is achieved due to each of the phases of the design stage along with the adjustment process of the exploitation stage. As a result, it can be said that the proposed design process makes it possible to obtain a reliable system, which, in addition, has been validated by the expert during its construction in the labeling phase. Furthermore, this small total error rate also demonstrates that the clustering phase has made a successful classification finding the proper number of clusters. Note that an error in the number of clusters would have led to a very high error rate.

The number of clusters the system can identify is limited, and it is defined during the design phase. In particular, assuming that there can be a huge number of possible faults in a network, among all possible fault cases, only the most frequent will be identified. As a result, the diagnosis system will not find any cluster related to those faults that are not present in the training data set, or whose presence is scarce. Therefore, both the rare and new failures will not be considered by the system. This issue is an inherit limitation of the unsupervised systems. Consequently, if the diagnosis of more faults is required, the system should be redesigned using a larger volume of training data. However, as these data are unlabeled, it is not possible to ensure that this new training data set includes occurrences of those new failures. In addition, new KPIs may be required to have new information to facilitate the identification of new faults.

When analyzing the specific states that are wrongly diagnosed by the BMU phase, it can be observed that a majority of them activate border neurons. In particular, 92.5% of the cases misdiagnosed by the BMU phase are assigned to a border neuron. Therefore, this justifies the decision of proposing an adjustment focused on the borders of the clusters. With that unsupervised adjustment, 24.3% of those cases are satisfactorily corrected. This has reduced the total error rate from 1% to 0.77%. It should be noticed that even a little improvement in the percentage, given the high number of cells in the network, means a considerable improvement in the number of cells correctly diagnosed. Furthermore, the use of the silhouette index provides a confirmation of the obtained diagnosis in the most difficult cases.

D. Algorithm Complexity Evaluation

Here, the complexity of the iterative procedures is discussed. In particular, the proposed diagnosis system is composed of several iterative algorithms in its design stage: both the rough-tuning and the fine-tuning of the training phase as well as the proposed unsupervised clustering. The rough-tuning phase is performed by the Batch algorithm, which has computational complexity of approximately $\mathrm{O}(\mathrm{P} \cdot \mathrm{J} \cdot \mathrm{M})/2$ according to [34]. It is recalled that P is the total number of cases in the training data set, J is the number of neurons, and M is the dimension of the input vector. To achieve convergence in the rough-tuning phase, several thousand iterations of this algorithm are required. However, the computational complexity of the sequential algorithm in the fine-tuning phase is about double of the Batch algorithm, that is, $\mathrm{O}(\mathrm{P} \cdot \mathrm{J} \cdot \mathrm{M})$ [34]. Furthermore, this phase requires more iterations to converge, i.e., several thousand iterations. Regarding the unsupervised clustering, which consists of a combination of mechanisms, its computational complexity is determined by the upper level, i.e., Ward's hierarchical method whose computational complexity is at least $\mathrm{O}(\mathrm{J}^2)$ [23]. However, it is executed over a few iterations determined by the maximum number of identifiable clusters (TC).

The running time of the procedures is strongly dependent on the number of iterations that each procedure is executed. Thus, the running time of the training phases varies according to the configured training length. Table VIII presents the running time of each iterative process calculated when the diagnosis system was designed with the simulated training data set, whose training length (i.e., the number of iterations of each procedure) is presented in Table III. As previously stated, the evaluation of the system has been performed with only one validation data set. Furthermore, these experiments were conducted on an Intel Core i5-2540M at 2.60 GHz and 4-GB memory. The operating system was windows 7 Enterprise. As it can be observed, the fine-tuning phase is the most time-consuming part. It should be stressed that the duration of the design phase is not as critical as the duration of the exploitation stage, which should identify the problem as fast as possible to minimize the time-to-resolution. For the proposed diagnosis system, the execution of the exploitation stage is instantaneous.

TABLE VIII
RUNNING TIME [S] OF EACH PROCEDURE

Rough-tuning	Fine-tuning	Unsupervised clustering
2.40	681.39	0.33

TABLE IX
PARAMETERS OF THE REAL LTE NETWORK

Parameter	Configuration
Network Layout	Urban area
Number of cells	8679
System bandwidth	10 MHz
Number of PRBs	50
Frequency reuse factor	1
Max. Transmit power	46 dBm
Maximum Tx Power of UE	23 dBm
Horizontal HPBW	65 °
HOM	3 dB
KPI time period	Hourly
Number of observed cells	45
Number of days under observation	6 days (on average)
Size of the training dataset	14478 unlabelled cases

E. Demonstration in a Live Network

Once the performance of the proposed diagnosis system has been assessed with simulated data, it has been applied in a live LTE network to demonstrate its usability and effectiveness using real and unlabeled data.

1) Details of the Analyzed Live LTE Network: The analysis of the proposed system has been conducted in a real LTE network of a big urban area corresponding to a city with a population of nearly four million. This LTE network has been chosen because its deployment is extensive and well established. The characteristics of this LTE network (such as transmission power, handover parameters, system bandwidth, etc.) are summarized in Table IX. It consists of more than 8000 different cells; hence, there is a great variety of cells, each of them located at different locations suffering different environment conditions. To obtain a big training data set, 45 cells have been randomly chosen among all the available cells in the network. From those cells, the values of the KPIs have been stored at an hourly level during an observation period of six days (on average). As a result, a training data set with a total of 14 478 different unlabeled cases has been obtained. It should be noted that it is important to store the state of the same cell at different hours because the values of the KPIs are very dependent on traffic conditions and the volume of served users, which vary with time. Therefore, several cases of the same cells have been stored over time, instead of storing a single case of different cells for the same hour. This ensures that the training data set includes cases that are affected by the traffic variations.

In particular, the KPIs selected to perform the diagnosis are some of the most common KPIs used by the experts in their manual troubleshooting tasks. Furthermore, these KPIs are associated with the main categories in mobile networks: connectivity, e.g., accessibility, retainability, and failed radio resource control (RRC) connection rate; mobility, e.g., HOSR, number of ping-pong HO and Inter-Radio Access Technology (IRAT) HO Rate; quality, e.g., number of bad coverage report and average of the Received Strength Signal Indicator (RSSI); capacity, e.g., number of RRC connections and load average of CPU; and, configuration, e.g., antenna tilt.

- Accessibility: This shows the ability of the cell to provide the service requested by the user under acceptable conditions [35]. Therefore, it is usually used to identify the percentage of connections that have got access to that cell over the KPI time period. As a result, a low value of accessibility shows that there are many blocked connections.
- Retainability: This KPI is the same as that described in Section V-A. That is, it represents the percentage of connections that are not interrupted or ended prematurely out of the total number of connections [35]. Thus, a high value of retainability determines that a majority of the connections have been successfully finished.
- Failed RRC Connections Rate: A successful RRC connection [36] determines that a user has been provided with the LTE resources required to transfer any kind of data. Thus, this KPI determines the ratio between the total number of failed RRC connections and the total number of requested RRC connections.
- HOSR: As stated in Section V-A, these KPIs show how well a cell performs the handover functionality providing a satisfactory mobility to their users given that it represents the number of HOs that have been successfully performed over the total number of HOs (considering both successful and failed HOs) [35].
- Number of ping-pong HO: This KPI counts the total number of ping-pong HOs that happen during the KPI time period. A ping-pong HO occurs when the user equipment (UE) switches between two cells repeatedly in a short time period [37]. This KPI is considered given that the ping-pong HO is a critical issue on the HO procedure that negatively affects the performance of a cell.
- IRAT HO Rate: An inter-radio access technology HO is a mobility process whereby users switch their connections from one RAT to another. In this case, this KPI represents the percentage of users in LTE that perform an IRAT HO from LTE to a different RAT over the total number of connections successfully finished. A high IRAT HO rate means that a lot of users are leaving LTE.
- Number of bad coverage report: It counts the number of signal level measurements in which Event A2 [36] of the mobility process is met, that is, the total number of times the received signal level from the serving cell is under an absolute threshold. A high value of this KPI gives an indication of a lack of coverage.
- Average RSSI: RSSI is the wide band power received by the user considering both the desired signals and the rest of received power due to thermal noise, adjacent channel interference, etc. Therefore, this KPI is calculated as the average of all the RSSI reported over the KPI time period.
- Number of RRC connections: This is the number of RRC connection attempts that have been successfully established. This KPI is a measure of the amount of users served by the cell.
- Average CPU Load: This is the weighted average of the CPU processes over the KPI time period. A cell with a high average load means that it has overload problems.
- Tilt: This is the antenna configuration parameter that determines the angle that the antenna forms with the horizontal plane. This means that the smaller the antenna tilt, the higher its coverage area.

2) Construction of the Proposed Diagnosis System: According to the proposed design process, the diagnosis system has been built using the real training data set previously presented. The first point to mention is that this training data set is much larger than the artificial training data set (i.e., 14 478 real cases against 550 simulated cases). This would result in an excessive rise of the running time of the training procedure, which is the most critical. As a result, the first design decision was to reduce the training length of the fine-tuning phase to 10% of the training length used with the artificial data set. The rest of the configuration parameters were set up with the same configuration. Once the training and clustering phases were completed, during the labeling phase, it was found that the obtained classification was fragmented. Therefore, as explained in Section IV, the training and clustering phases were repeated with different configuration parameters. To that end, the final neighborhood radius of the rough-tuning phase and the initial learning rate of the fine-tuning phase were reduced to do the training with more resolution. The particular values used for those training parameters are shown in Table III. Furthermore, the maximum number of clusters identifiable by the system remains 10. With this design configuration, four statistically different clusters have been found by the system. In addition, all clusters are constituted by adjoining neurons, which determines that both training and clustering phases have been successful. To label each of them, their statistical behavior has been analyzed through the pdfs of the KPI estimated by the kernel smoothing function, as previously stated in this paper. Nevertheless, here, the mean and the standard deviation of each KPI given each cluster have been presented in Table X instead of the figures of those pdfs because of space constraints.

First, the statistical behavior of the clusters is analyzed to find the normal one. This will be the cluster whose KPIs have the most common and less-deteriorated values. On this basis, cluster 3 has been labeled as Normal (see Table XI), given the

TABLE X
MEAN AND STANDARD DEVIATION OF EACH KPI GIVEN EACH CLUSTER

KPI	Mean				Standard deviation			
	C 1	C 2	C 3	C 4	C 1	C 2	C 3	C 4
Accessibility [%]	76.04	99.07	99.78	0.12	25.79	7.01	0.34	3.47
Retainability [%]	86.88	99.11	99.75	0	23.36	6.96	0.36	0
Failed RRC Connections Rate [%]	18.40	0.41	0.23	0	16.80	0.89	0.41	0
HOSR [%]	83.24	97.77	96.88	6.82	16.79	8.62	12.65	25.18
Number of ping-pong HO	4040	36.98	34.02	0.15	4363	34.14	146.51	1.31
IRAT HO Rate [%]	1.38	3.47	1.16	0.07	1.32	5.23	1.17	0.53
Number of bad coverage report	1241	1352	198	0.35	1041	903.28	262.55	2.12
Average RSSI [dBm]	-107.73	-113.82	-115.86	-15.22	4.39	2.27	2.72	39.68
Average CPU Load	42.32	32.69	21.66	22.65	12.17	9.74	3.53	4.17
Number of RRC connections	28867	13990	5467	0.04	17943	6459	4474	1.06
Tilt [°]	0.63	1.38	2.34	3.63	1.13	1.67	2.40	3.48

TABLE XI
LABEL ASSIGNED TO EACH CLUSTER

Clusters	Label
Cluster 1	Overload
Cluster 2	Lack of Coverage
Cluster 3	Normal
Cluster 4	Non-operating cell

following reasons. This cluster represents cells whose connectivity procedure works successfully given that it has high accessibility and retainability (both KPIs are around 99%) along with a very low failed RRC connection rate. The HOSR is adequate, and both the absolute number of ping-pong HOs and the IRAT HO rate are relatively low (around 34.02 and 1.16%, respectively); this means that there are no problems in the mobility process. Regarding the quality, the low number of bad coverage reports determines that the served users measure the cell with a high signal level. Furthermore, the average RSSI is around -115.86 dBm; hence, it is within the desired range from -120 to -114 dBm. Finally, the low load average indicates that there is no overload. As a result, this cluster presents a normal performance.

By comparing cluster 1 with the Normal cluster, it can be seen that all its KPIs are deteriorated. In particular, the low value of accessibility and the high failed RRC connection rate indicate that there are a lot of users that cannot establish a connection. In addition, there are a large number of dropped connections, as the low values of retainability show (86.88% on average). These symptoms, along with the high average CPU load (around 42.32), reveal that this cluster matches cells that have overload problems. Moreover, it is fully in line with the rest of the symptoms such as the high number of RRC connections. Furthermore, as the number of users increases, the interference levels suffered by the users are incremented, causing a significant deterioration of the average RSSI (which is around -107 dBm, outside the acceptable range). In conclusion, the cell is overloaded due to the high amount of traffic causing that the cells cannot maintain the service under acceptable conditions and further blocks the connections of new users.

Regarding cluster 2, its accessibility, retainability, and failed RRC connection rate have a good statistical behavior (similar to the normal one). Furthermore, the KPIs related to the HO procedure also present normal values, except the IRAT HO rate, which is higher than normal. This indicates that a lot of users are leaving the LTE technology, which is undesirable. By analyzing the rest of KPIs, it can be observed that, in addition, both the number of bad coverage report and the average RSSI are deteriorated. In this case, the high number of established RRC connections, along with the low values of the antenna tilt, suggests that the cells are covering a higher than necessary coverage area, and thus, they are serving users with bad signal conditions. On this basis, it is concluded that this behavior is in line with the symptoms of lack of coverage.

Finally, concerning cluster 4, a majority of its KPIs are concentrated near zero. Both accessibility and retainability are practically zero, which means that there is hardly any connection established in those cells. As a result, this cluster is labeled as a nonoperating cell.

3) Case Study: Diagnosing Problematic Cells Over Time: To demonstrate the performance of the automatic diagnosis system and validate the assigned labels, the system has been applied to diagnose different cells of this real LTE network.

The first chosen cell is a problematic cell that has been manually reconfigured over time to improve its performance. Therefore, for this study, it has been analyzed by the proposed diagnosis system during the days when the troubleshooting tasks were being carried out. Fig. 6 shows the evolution of the number of bad coverage reports (presented by a continuous black line), since it is the most relevant KPI in this situation. Furthermore, the diagnosis achieved with the proposed system over the same period of time is superimposed. In particular, the diagnosis automatically obtained during the first three days determines that this cell is not operating. This matches the troubleshooting tasks, which reveal that this cell was inactive during this period of time. After the cell was launched, its KPIs indicate that the performance of the cell was deteriorated. In particular, the diagnosis varies between normal and lack of coverage problem over time and in accordance with traffic. Therefore, only during the busy hours does the existing problem comes to light. This diagnosis is in line with the manual diagnosis performed by the operator who decided to increment the tilt from 0° to 6° to control the radio frequency conditions and improve the overall performance of the cell. This change is represented in Fig. 6 by a vertical line, determining the instant in which the antenna tilt was changed. Analyzing this figure, it can be seen that this change fixed the problem, reducing the number of bad coverage reports in that cell. In accordance with this, after the change, the cell is automatically diagnosed as normal by the system.

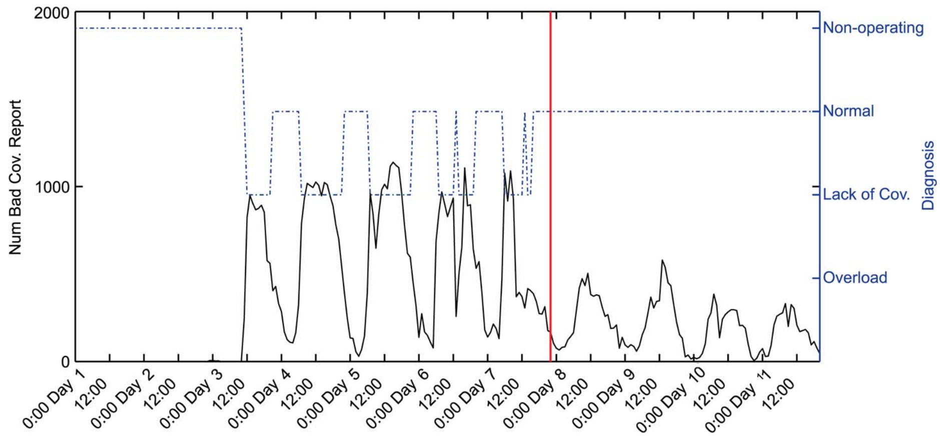

Fig. 6. Number of bad coverage report values of the diagnosed cell along with the obtained diagnosis.

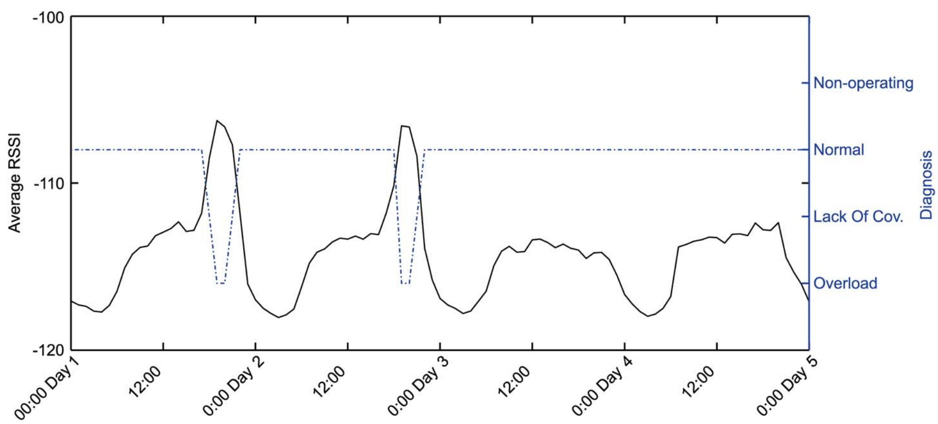

Fig. 7. Average RSSI values of the diagnosed cell along with the obtained diagnosis.

To validate the overload cluster, a different problematic cell is analyzed. In this case, the diagnosis system determines that the cell presented overload problems on two occasions. This is reflected in all the KPIs whose values are extremely deteriorated. As an example of the latter, the average RSSI is shown along with its diagnosis in Fig. 7. It can be seen that the overload problem matches the hours in which the values of the average RSSI is deteriorated. According to the information provided by a troubleshooting expert, the deterioration of these KPIs is due to the high connection attempts caused by peak traffic.

VI. Conclusion

An automatic diagnosis system as part of a self-healing network has been proposed in this paper. This system is built through unsupervised techniques with the aim of obtaining a system that represents the normal and faulty behaviors of the real network. The use of unsupervised techniques guarantees that the system can be built without historical reports of solved cases while simultaneously enabling the system to identify new faults that are not previously known. Even so, the clusters derived from the proposed system are labeled by an expert based on their statistical behavior, although the effort required from experts is negligible compared with that required for supervised methods. In particular, the pdfs of each KPI given each cluster are estimated, taking into account all the cases in the training data set, which provides more information that only considering the weight vectors of the neurons. By performing supervised labeling, experts can detect errors in the clustering, identify the behavior of each cluster, assign the best suited fault cause to each cluster based on their knowledge, and verify whether the system is right or not. As a result, this stage is not only a labeling phase but also a validation phase.

The main requirement is that the identification process must be relatively prompt, objective, and automatic. The key element for achieving this is the proposed adjustment phase. To avoid slowing down the exploitation process, this technique only acts when the traditional mapping is more likely to be wrong, that is, when the activated neuron is a border neuron between two or more clusters. Furthermore, this phase attempts to correct the errors in an objective and automatic way. This correction is done based on the Xth percentile of all distances between the input and each cluster and the evaluation provided by the average silhouette index.

To assess the proposed approach, the diagnosis system has been built with both simulated and real data, showing how the construction phase must be done and how the diagnosis is performed in a live network. The obtained results demonstrate the value of the integrated approach. Furthermore, the proposed diagnosis system has been compared with reference mechanisms to objectively evaluate its effectiveness. It is important to point out that the proposed diagnosis system is highly accurate, taking into account that it has been built using unsupervised techniques. Finally, it can be concluded that this system could be part of a self-healing network where specific corrective actions are taken after the automatic diagnosis stage.

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